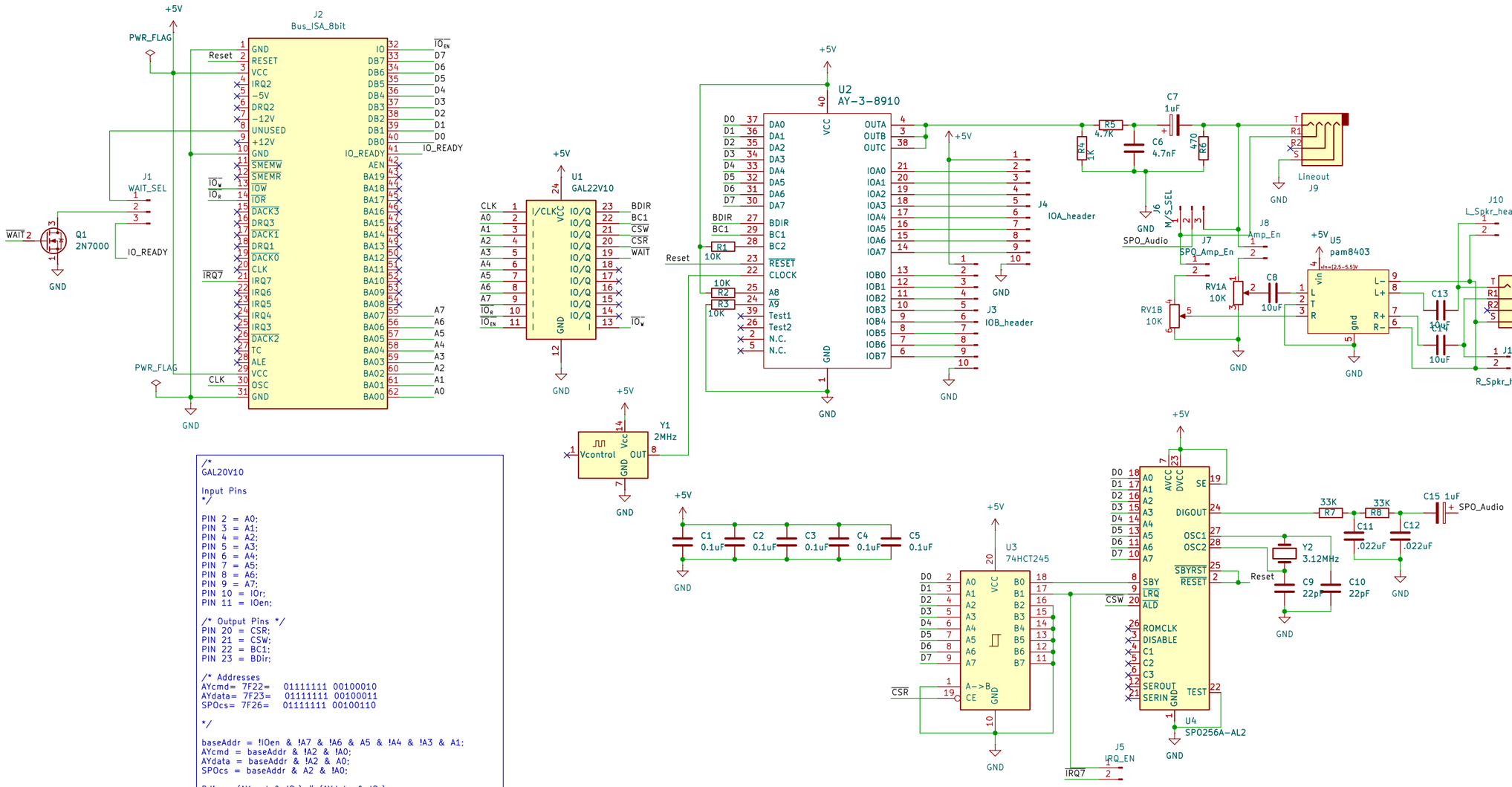



```
/*
GAL20V10

Input Pins
*/

PIN 2 = A0;
PIN 3 = A1;
PIN 4 = A2;
PIN 5 = A3;
PIN 6 = A4;
PIN 7 = A5;
PIN 8 = A6;
PIN 9 = A7;
PIN 10 = !Or;
PIN 11 = !Oen;

/* Output Pins */
PIN 20 = CSR;
PIN 21 = CSW;
PIN 22 = BC1;
PIN 23 = BDir;

/* Addresses
AYcmd= 7F22=   01111111 00100010
AYdata= 7F23=   01111111 00100011
SPOcs= 7F26=   01111111 00100110

*/

baseAddr = !IOen & !A7 & !A6 & A5 & !A4 & !A3 & A1;
AYcmd = baseAddr & !A2 & !A0;
AYdata = baseAddr & !A2 & A0;
SPOcs = baseAddr & A2 & !A0;

Bdir = (AYcmd & !Or) # (AYdata & !Or);
BC1 = (AYdata & !!Or) # (AYcmd & !Or);
CSR = !SPOcs # !Or
CSW = !SPOcs # !!Or

/* AYcmd   AYdata   !Or | Bdir  BC1   | State
   0     0     X   |  0  0  |  INACTIVE
   0     1     0   |  0  1  |  PSG Read
   0     1     1   |  1  0  |  PSG Write
   1     0     1   |  1  1  |  Address Latch

/CSW = SPOcs & !Or = Write to SPO
/CSR = SPOcs & !!Or   = Read SPO status
*/
```

| | | |
|---|---|---|
| Sheet: / | | |
| File: ISA_AY-3-8910_Sound.kicad_sch | | |
| **Title:** | | |
| Size: A3 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.1 | | Id: 1/1 |